

2021

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

(Compulsory)

1. Pick up the correct alternative for each of
the following questions : $10 \times 2 = 20$

(a) What is the ready state of a process ?

- (i) When process is scheduled to run
after some execution

- (ii) When process is unable to run until source task has been completed
 - (iii) When process is using the CPU
 - (iv) None of the mentioned
- (b) What is a short term scheduler ?
- (i) It selects which process has to be brought into the ready queue
 - (ii) It select which process has to be executed next and allocates CPU
 - (iii) It selects which process has to be removed memory from swapping
 - (iv) None of the mentioned
- (c) Round Robin Scheduling falls under the category of :
- (i) Non preemptive
 - (ii) Preemptive

- (iii) Both (i) & (ii)
 - (iv) None of the above
- (d) A minimum of ____ variable is/are required to be shared between processes to solve the critical section problem :
- (i) One
 - (ii) Two
 - (iii) Three
 - (iv) Four
- (e) Binding of instructions and data to memory address can be done at :
- (i) Compile time
 - (ii) Load time
 - (iii) Execution time
 - (iv) All of the above
- (f) Real time system must have :
- (i) Preemptive kernels
 - (ii) Non preemptive kernels

- (iii) Both (i) & (ii)
- (iv) None of the above
- (g) In Unix each process is identified by its :
 - (i) PCB
 - (ii) Device queue
 - (iii) Process identifies
 - (iv) None of the above
- (h) CPU scheduling is the basis of :
 - (i) Multiprocessor system
 - (ii) Multiprogramming system
 - (iii) Large memory size system
 - (iv) None
- (i) In OS which of the following is/are CPU scheduling algorithm :
 - (i) Priority
 - (ii) Round Robin

- (iii) SJF
 - (iv) All
- (j) To access the services of the OS the interface is provided by the ____.
- (i) Library
 - (ii) System calls
 - (iii) Assembly instruction
 - (iv) All

Section-B (1-Ques. left)

Answer any **four** questions : 4×5=20

- ✓ 2. What is throughput, turnaround time, waiting time and response time. left*
- ✓ 3. Define time sharing operating system.
4. What do you mean by context switching ?
5. Explain virtual memory and its uses.
6. What are the various layers of file system ?
- 7. Differentiate between internal and external fragmentation.

✓ 8. What are the various criteria for CPU scheduling ?

9. Write short notes on firewall.

Section-C (1 Ques. / 44)

Answer any **two** questions of the following :

$$2 \times 15 = 30$$

✓ 10. What is an operating system ? What is the need for an operating system ? Discuss the major function of an OS.

✓ 11. Consider the following set of process with the length of CPU burst time given in millisecond

Process	Burst time (msec)	Arrival time (m Sec)
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P_1	24	0
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P_2	7	3
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P_3	6	5
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P_4	10	10
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Draw Gantt chart for FCFS, SJF and RR (quantum⁴) scheduling algorithms. Calculate the average waiting time and turn around time for each of the above mentioned algorithms.

12. What are the various threats to security of a system ? Elaborate in details the different mechanism of protection and security.

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